

DaSEH Day 7 Cheatsheet

Data Visualization

Major concepts

- **tidyverse (ggplot2):** Produces graphics.
- **Tidy data:** Each variable is a column, and each observation is a row.
 - Having data in tidy format and long format makes creating plots easier!
- **Plot layers:** Use + at the end of each line when adding layers to plots.
- **Pipes:** |> and %>% do not add plot layers, but they can pipe data into plots.
 - **Works:** `ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_boxplot()`
 - **Works:** `iris |> ggplot(aes(x = Species, y = Petal.Length)) + geom_boxplot()`
 - **Does not work:** `iris |> ggplot(aes(x = Species, y = Petal.Length)) |> geom_plot()`

Functions

Library/Package	Piece of code	Example of usage	What it does
tidyverse (ggplot2)	<code>ggplot()</code>	<code>ggplot(data = iris)</code>	Begins a plot that is finished by adding layers.
tidyverse (ggplot2)	<code>aes()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length))</code>	Maps variables to visual properties in a <code>ggplot()</code> object.
tidyverse (ggplot2)	<code>geom_boxplot()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_boxplot()</code>	Creates a boxplot when added as a layer to a <code>ggplot()</code> object.
tidyverse (ggplot2)	<code>geom_density()</code>	<code>ggplot(data = iris, aes(x = Petal.Length)) + geom_density()</code>	Creates a smoothed density plot when added as a layer to a <code>ggplot()</code> object.

Library/Package	Piece of code	Example of usage	What it does
tidyverse (ggplot2)	<code>geom_point()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_point()</code>	Creates a scatterplot when added as a layer to a <code>ggplot()</code> object.
tidyverse (ggplot2)	<code>geom_line()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_line()</code>	Creates a line plot by connecting points in order of the x-axis variable.
tidyverse (ggplot2)	<code>geom_hline()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_point() + geom_hline(yintercept = 4)</code>	Adds horizontal lines to a plot.
tidyverse (ggplot2)	<code>theme_classic()</code>	<code>ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + theme_classic()</code>	Displays a <code>ggplot()</code> object without grid lines. See the ggtheme documentation for additional themes.
tidyverse (ggplot2)	<code>xlab(); ylab(); ggtitle()</code>	<code>ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + xlab("Petal Length")</code>	Modifies the x-axis label, y-axis label, and plot title, respectively.
tidyverse (ggplot2)	<code>facet_grid()</code>	<code>ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + facet_grid(~Species)</code>	Creates a grid of plots using specified variables to subset the data.
tidyverse (ggplot2)	<code>facet_wrap()</code>	<code>ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + facet_wrap(~Species, scales = "free", nrow = 2)</code>	Creates individual plots using specified variables to subset the data, with flexible layouts and axis scales.
tidyverse (ggplot2)	<code>ggsave()</code>	<code>ggsave(filename = "plotname.pdf")</code>	Saves the last plot in the working directory.

NOTE: Click [here](#) for a summary of the `ggplot2` theme system.

Factors

Major concepts

- **Factor:** A special character vector whose elements have predefined groups, or levels. Factors are useful for qualitative or categorical variables, such as grade levels. Specify factors with the `factor` function so R recognizes them correctly.

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R	<code>factor()</code>	<code>fact_qual <- factor(c("poor", "fine", "good"))</code>	Creates a factor from a vector. Levels are ordered alphanumerically by default.
Base R	<code>levels()</code>	<code>levels(fact_qual)</code>	Shows the levels and order of a factor vector.
tidyverse (forcats)	<code>as_factor()</code>	<code>fact_qual <- as_factor(c("poor", "fine", "good"))</code>	Creates a factor from a vector. Levels are ordered by first appearance in the data by default.
tidyverse (forcats)	<code>fct_reorder()</code>	<code>ggplot(iris, aes(x = fct_reorder(Species, Sepal.Width, mean), y = Sepal.Width)) + geom_boxplot()</code>	Reorders factor levels according to another variable. Here, <code>Species</code> is ordered by the mean of <code>Sepal.Width</code> .

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.